

COSTING

Introduction

One of the primary objectives of the Varsity Cyber Clash E-Sports Tournament is to provide a reliable and high-performance networking environment while maintaining reasonable operational costs. The proposed budget covers hardware procurement, network infrastructure installation, electrical setup, software licensing, internet connectivity, and maintenance services required to support approximately 200 participants, tournament officials, broadcasters, and spectators.

The proposed infrastructure is designed to ensure low network latency, stable connectivity, efficient tournament management, secure data handling, and uninterrupted online gameplay throughout the event.

Hardware Procurement Cost

Equipment	Quantity	Cost (RM)
Wireless AC1200 Gigabit Router	1	200
D-Link 24-Port Gigabit Switch	2	774
TP-Link Wireless Access Point	6	1,782
Dell Vostro 3681 Workstation	30	47,520
Dell SE2426H 24-inch Monitor	100	43,900
Dell PowerEdge T620 Server	6	36,942
Alienware Aurora Gaming Desktop	70	1,378,930
Layer 3 Core Switch (S5800-48F4SR)	1	7,405
TP-Link TL-SG3452 Managed Switch	2	4,398

Total Hardware Procurement Cost

Network Infrastructure Procurement Cost

Item	Quantity	Unit Cost (RM)	Total Cost (RM)
Cat5e UTP Cable (305m Box)	2	300	600
RJ45 Connectors (100 pcs)	2 Pack	30	60
Patch Panel 24-Port	2	150	300
Wall Faceplate & Keystone Jack	20	15	300
Cable Trunking & Accessories	1 Lot	400	400

Total Network Infrastructure Cost

RM 1,660

Electrical Setup Cost

Item	Quantity	Cost (RM)
Power Extension Sockets	20	600
UPS Backup Power Supply	2	2,000
Circuit Protection Equipment	1 Lot	500

Total Electrical Setup Cost

RM 3,100

Internet Subscription Cost

A dedicated high-speed internet connection is required to ensure stable gameplay, tournament administration, livestream broadcasting, and communication services.

Service	Monthly Cost (RM)
500 Mbps Fiber Internet Package	199

Annual Internet Cost

RM199 × 12 months = **RM 2,388**

Software Licensing Cost

Software	Quantity	Cost (RM)
Windows 11 Pro License	100	10,000
Antivirus Solution	100	10,000

Total Software Licensing Cost

RM 20,000

Manpower and Maintenance Cost

Service	Annual Cost (RM)
Network Installation Technician	6,500
Preventive Maintenance	3,500
Troubleshooting Support	3,000

Total Manpower and Maintenance Cost

RM 13,000

Overall Project Cost Summary

Category	Cost (RM)
Hardware Procurement	1,521,851
Network Infrastructure Procurement	1,660
Electrical Setup	3,100

Software Licensing	20,000
Internet Subscription (Annual)	2,388
Manpower & Maintenance	13,000

Grand Total Project Cost

RM 1,561,999

Cost Justification

The majority of the project budget is allocated to gaming desktops, servers, and network infrastructure. The gaming desktops provide the necessary processing power and graphics performance required for competitive E-Sports tournaments. Dedicated servers are deployed to host tournament management systems, registration services, game servers, FTP services, and live streaming applications.

Network devices such as managed switches, access points, and Layer 3 core switches ensure efficient traffic management and low-latency communication between tournament participants. Additional expenditure is allocated to software licensing, internet connectivity, electrical safety equipment, and maintenance services to ensure uninterrupted event operations.

The proposed infrastructure provides a scalable, secure, and reliable networking environment capable of supporting more than 200 participants and spectators during the Varsity Cyber Clash E-Sports Tournament.